

Tutoring Session
 Tuesday 9:00pm - 11:00pm
 engineering Building South 426

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ u|_{x=0} = u_0 \quad u|_{x=a} = u_0 \left[3\left(\frac{y}{b}\right)^2 - 2\left(\frac{y}{b}\right)^3 \right] \\ u_y|_{y=0} = 0 \quad u_y|_{y=b} = 0 \end{cases}$$

$$u(x, y) = X(x) Y(y)$$

$$X''Y + XY'' = 0$$

$$\frac{X''}{X} = -\frac{Y''}{Y} = \lambda$$

$$\begin{cases} X'' - \lambda X = 0 \\ Y'' + \lambda Y = 0 \end{cases} \quad Y'(0)|_{y=0} = 0 \quad Y'(b)|_{y=b} = 0$$

① ① $\lambda = 0 \quad Y = Ay + B$
 $Y = B_0$

② $\lambda < 0 \quad Y = Ae^{-\alpha y} + Be^{\alpha y} \quad \alpha = \sqrt{-\lambda}$
 $Y'(0) = -\alpha A + \alpha B = 0$
 $Y'(b) = -\alpha Ae^{-\alpha b} + \alpha Be^{\alpha b} = 0$
 $A = B = 0 \quad X$

③ $\lambda > 0 \quad Y = A \cos \sqrt{\lambda} y + B \sin \sqrt{\lambda} y$
 $Y' = -\sqrt{\lambda} A \sin \sqrt{\lambda} y + B \sqrt{\lambda} \cos \sqrt{\lambda} y$
 $Y'(0) = 0 \Rightarrow B = 0$
 $Y'(b) = 0 \Rightarrow -\sqrt{\lambda} A \sin \sqrt{\lambda} b = 0$
 $\sqrt{\lambda} = \frac{n\pi}{b} \quad n = 1, 2, 3, \dots$

$$\lambda = \left(\frac{n\pi}{b}\right)^2 \quad \lambda = \left(\frac{n\pi}{b}\right)^2$$

$$\begin{cases} X'' - \lambda X = 0 \\ X_n = C_n e^{\frac{n\pi}{b}x} + D_n e^{-\frac{n\pi}{b}x} \end{cases}$$

$$\lambda = 0$$

$$X'' = 0 \quad X_0 = C_0 x + D_0$$

$$u = \underbrace{(C_0 x + D_0) B_0}_{X_0 Y_0} + \sum_{n=1}^{\infty} \underbrace{(C_n e^{\frac{n\pi}{b}x} + D_n e^{-\frac{n\pi}{b}x})}_{X_n} \underbrace{\cos \frac{n\pi}{b} y}_{Y_n}$$

$$u|_{x=0} = u_0 \text{ constant}$$

$$= D_0 B_0 + \sum_{n=1}^{\infty} (C_n + D_n) \cos \frac{n\pi}{b} y$$

$$u|_{x=0} = u_0 \sim$$

$$= D_0 B_0 + \sum_{n=1}^{\infty} \frac{(C_n + D_n) \cos \frac{n\pi}{b} y}{C_n + D_n = 0} \star$$

$$\Rightarrow D_0 B_0 = u_0, \quad C_n + D_n = 0 \star$$

$$u|_{x=a} = u_0 \left[3 \left(\frac{y}{b} \right)^2 - 2 \left(\frac{y}{b} \right)^3 \right]$$

$$= \underline{B_0 C_0 \cdot a + u_0} + \sum_{n=1}^{\infty} \left(\underline{C_n e^{\frac{n\pi}{b} a} + D_n e^{-\frac{n\pi}{b} a}} \right) \cos \frac{n\pi}{b} y$$

$$\star C_n e^{\frac{n\pi}{b} a} + D_n e^{-\frac{n\pi}{b} a} = \frac{2}{b} \int_0^b \cos \frac{n\pi}{b} y \cdot u_0 \left[3 \left(\frac{y}{b} \right)^2 - 2 \left(\frac{y}{b} \right)^3 \right] dy$$

$$\underline{B_0 C_0 \cdot a + u_0} = \frac{1}{b} \int_0^b u_0 \left[3 \left(\frac{y}{b} \right)^2 - 2 \left(\frac{y}{b} \right)^3 \right] dy$$

b

$$u_{tt} - a^2 u_{xx} = b x(l-x)$$

$$V(x, t) = \frac{b}{12a^2} x(x^3 - 2lx^2 + l^3) \checkmark$$

$$V|_{x=0} = 0 \quad V|_{x=l} = 0$$

$$V_{tt} = 0$$

$$a^2 \frac{b}{12a^2} \left(\frac{x^4 - 2lx^3 + l^3 x}{V_{tt} - a^2 u_{xx}} \right)'' = 12x^2 - 12lx$$

$$= b x(l-x)$$

$$V(x) = p_4(x) = \frac{a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0}{p_4(0) = p_4(l) = 0}$$

$$\begin{cases} -a^2 u_{xx} = b x(l-x) \\ V(0) = V(l) = 0 \end{cases}$$

$$\text{let } w = u - V$$

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0 \\ w|_{x=0, l} = 0 \\ w|_{t=0} = -\frac{b}{12a^2} x(x^3 - 2lx^2 + l^3) \quad \varphi(x) \\ w_t|_{t=0} = 0 \quad \psi(x) \end{cases}$$

$$\lim_{t \rightarrow 0} w_t = 0$$

$$\psi(x)$$

$$\nabla^2 u = -2$$

$$u_{xx} + u_{yy} = -2$$

$$v_{xx} + v_{yy} = -2 \quad v|_{x=0, a} = 0$$

$$v(x) = Ax^2 + Bx + C$$

$$v_{xx} + u_{yy} = 2A = -2$$

$$A = -1, \quad C = 0$$

$$-x^2 + Bx|_{x=a} = 0$$

$$B = a$$

$$v(x) = -x^2 + ax$$

$$w = u - v$$

$$u - \tilde{v}$$

$$\begin{cases} w_{xx} + w_{yy} = 0 \\ w|_{x=0, a} = 0 \\ w|_{y=b/2} = -(-x^2 + ax) \\ w|_{y=-b/2} = -(-x^2 + ax) \end{cases}$$

$$\text{let } w = x \cdot Y$$

$$\nabla^2 u = x^2 y$$

$$u_{xx} + u_{yy} = x^2 y$$

$$\underline{x^2 y} \quad \left(\frac{1}{12} x^4 + A x \right) y$$

$$\left(\frac{1}{12} a^4 + A - a \right) = 0 \quad A = \frac{-1}{12} a^3$$

$$V(x, y) = \left(\frac{1}{12} x^4 - \frac{1}{12} a^3 x \right) y \quad \star$$

$$W = U - V$$

$$\S \quad \begin{cases} U_{tt} - a^2 U_{xx} = 0 \\ U|_{x=0} = \cos \frac{\pi}{l} a t \quad U_x|_{x=l} = 0 \quad \checkmark \end{cases}$$

$$V(x, t) = \underline{\cos \frac{\pi}{l} x \cos \frac{\pi}{l} a t}$$

$$V_{tt} - a^2 V_{xx} = 0$$

$$\text{let } W = U - V$$

$$\begin{aligned} \S \quad & U|_{x=0} = A e^{-\alpha^2 k t} \\ & U|_{x=l} = B e^{-\beta^2 k t} \\ & V(x, t) = A f(x) \cdot e^{-\alpha^2 k t} + B g(x) e^{-\beta^2 k t} \\ & \left. \begin{aligned} f(0) &= 1 & g(0) &= 0 \\ f(l) &= 0 & g(l) &= 1 \end{aligned} \right\} \end{aligned}$$

$$V_t - k V_{xx} = 0$$

$$f(x) = \frac{\sin \alpha (l-x)}{\sin \alpha l}$$

$$g(x) = \frac{\sin \beta x}{\sin \beta l}$$

Problem 1, 2
(4)X

$$u_{xx} - 2u_{xy} - 3u_{yy} = 0$$

$$(\mathcal{D}_x^2 - 2\mathcal{D}_x\mathcal{D}_y - 3\mathcal{D}_y^2)u = 0$$

$$(\mathcal{D}_x - 3\mathcal{D}_y)(\mathcal{D}_x + \mathcal{D}_y)u = 0$$

$$(\mathcal{D}_x + \mathcal{D}_y)u = 0 \Rightarrow \phi_1(x-y)$$

$$(\mathcal{D}_x - 3\mathcal{D}_y)u = 0 \Rightarrow \phi_2(y+3x)$$

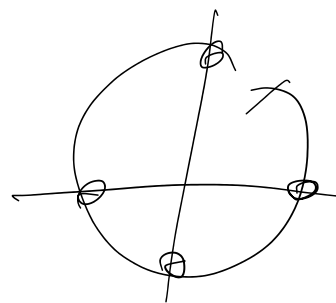
$$(\mathcal{D}_x - \lambda\mathcal{D}_y)u = 0 \Rightarrow \phi(y + \lambda x)$$

$$\phi'(y + \lambda x) \cdot \lambda - \lambda \phi'(y + \lambda x) = 0$$

$$u = \phi_1(x-y) + \phi_2(y+3x)$$

$$u_{xxxx} - u_{yyyy} = 0$$

$$(\mathcal{D}_x^4 - \mathcal{D}_y^4)u = 0$$



$$(\mathcal{D}_x - \mathcal{D}_y)(\mathcal{D}_x + \mathcal{D}_y)(\mathcal{D}_x - i\mathcal{D}_y)(\mathcal{D}_x + i\mathcal{D}_y)u = 0$$

$$u = \phi_1(x+y) + \phi_2(y-x) + \phi_3(y+ix) + \phi_4(y-ix)$$

$$2// \quad u_{xx} - u_{yy} = xy - x //$$

$$(D_x^2 - D_y^2)u = xy - x$$

$$u = \frac{1}{D_x^2 - D_y^2} (xy - x)$$

$$= \frac{1}{D_x^2} \underbrace{\frac{1}{1 - \frac{D_y^2}{D_x^2}}}_{\text{}} (xy - x)$$

$$\frac{1}{1-x} = 1 + x + x^2 + \dots$$

$$= \frac{1}{D_x^2} \left[1 + \frac{D_y^2}{D_x^2} + \frac{D_y^4}{D_x^4} \dots \right] (xy - x)$$

$$= \frac{1}{D_x^2} [xy - x + 0]$$

$$\frac{1}{D_x^2} D_y^2 (xy - x) = 0$$

$$= \frac{1}{D_x^2} (xy - x)$$

$$\frac{1}{D_x^2} x^2 = u$$

$$x^2 = D_x^2 \cdot u \quad u = \frac{1}{2} x^2$$

$$= \frac{1}{D_x^2} xy - \frac{1}{D_x^2} x$$

$$\left(\frac{1}{6} x^3 y - \frac{1}{6} x^3 \right)$$

$$+ \phi_1(y+x) + \phi_2(y-x)$$

$$u_{xx} - 2u_{xy} + u_{yy} = x^2 + y \quad \checkmark$$

$$(D_x^2 - 2D_x D_y + D_y^2)u = x^2 + y$$

$$(D_x - D_y)(D_x - D_y)u = x^2 + y$$

$$u = \frac{1}{(D_x - D_y)^2} x^2 + y$$

$$= \frac{1}{D_x^2} \frac{1}{(1 - \frac{D_y}{D_x})^2} x^2 + y$$

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1) x^n$$

$$= \frac{1}{D_x^2} \left[1 + 2 \frac{D_y}{D_x} + 3 \frac{D_y^2}{D_x^2} + \dots \right] (x^2 + y)$$

$$= \frac{1}{D_x^2} \left[x^2 + y + 2 \frac{D_y}{D_x} (x^2 + y) + \dots \right]$$

$$\frac{1}{D_x} D_y (x^2 + y)$$

$$\frac{1}{D_x} \cdot 1 = x$$

$$= \frac{1}{D_x^2} [x^2 + y + 2x]$$

$$= \frac{1}{12} x^4 + \frac{1}{2} x^2 y + 2 \cdot \frac{1}{6} x^3$$

$$\phi_1(y+x) + x \phi_1(y+x) + \dots$$

$$u_{xx} - 5u_{xy} + 6u_{yy} = x^2 + y$$

$$D_x^2 - 5D_x D_y + 6D_y^2 = x^2 + y$$

$$(D_x - 2D_y)(D_x - 3D_y)u = x^2 + y$$

$$u = \frac{1}{D_x - 2D_y} \frac{1}{D_x - 3D_y} x^2 + y$$

$$= \frac{1}{D_x^2} \left(\frac{1}{1 - 2 \frac{D_y}{D_x}} \right) \left(\frac{1}{1 - 3 \frac{D_y}{D_x}} \right) (x^2 + y)$$

$$\underbrace{\quad}_{\dots} \underbrace{\quad}_{D_x} \underbrace{\quad}_{D_x^2}$$

$$\frac{1}{1-x} = 1 + x + x^2 + \dots$$

$$\left(1 + 3 \frac{D_y}{D_x} + \frac{D_y^2}{D_x^2} + \dots\right) (x^2 + y)$$

$$= x^2 + y + 3 \frac{1}{D_x} D_y (x^2 + y) + 0$$

$$= x^2 + y + 3 \frac{1}{D_x} \cdot 1$$

$$= x^2 + y + 3x$$

$$\frac{1}{1-2 \frac{D_y}{D_x}} \cdot (x^2 + 3x + y) \quad \checkmark$$

$$= \left(1 + 2 \frac{D_y}{D_x} + \frac{2^2 D_y^2}{D_x^2} + \dots\right) (x^2 + 3x + y)$$

$$= x^2 + 3x + y$$

$$+ 2 \frac{1}{D_x} D_y (x^2 + 3x + y) \quad \frac{1}{D_x} \cdot 1$$

$$+ 4 \frac{1}{D_x^2} D_y^2 (x^2 + 3x + y) \quad \frac{1}{D_x^2} 0 = 0$$

$$= x^2 + 3x + y + 2 \cdot x$$

$$= x^2 + 5x + y$$

$$\frac{1}{D_x^2} (x^2 + 5x + y) = \frac{1}{12} x^4 + 5 \cdot \frac{1}{6} x^3 + \frac{1}{2} x^2 y$$

$$\phi_1(y+2x) + \phi_2(y+3x) + \quad \checkmark$$